

# Introduction to Machine Learning

Week 6

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1. (1 Mark) Entropy for a 90-10 split between two classes is:

- (a) 0.469
- (b) 0.195
- (c) 0.204
- (d) None of the above

**Sol.** (a)

2. (2 Mark) Consider a dataset with only one attribute(categorical). Suppose, there are 8 unordered values in this attribute, how many possible combinations are needed to find the best split-point for building the decision tree classifier?

- (a) 511
- (b) 1023
- (c) 512
- (d) 127

**Sol.** (d)

Suppose we have  $q$  unordered values; the total possible splits would be  $2^{q-1} - 1$ . Thus, in our case, it will be  $2^7 - 1 = 127$ .

3. (2 mark) Having built a decision tree, we are using reduced error pruning to reduce the size of the tree. We select a node to collapse. For this particular node, on the left branch, there are three training data points with the following outputs: 5, 7, 9.6, and for the right branch, there are four training data points with the following outputs: 8.7, 9.8, 10.5, 11. The average value of the outputs of data points denotes the response of a branch. The original responses for data points along the two branches (left & right respectively) were *response\_left* and, *response\_right* and the new response after collapsing the node is *response\_new*. What are the values for *response\_left*, *response\_right* and *response\_new* (numbers in the option are given in the same order)?

- (a) 9.6, 11, 10.4
- (b) 7.2; 10; 8.8
- (c) 5, 10.5, 15
- (d) Depends on the tree height.

**Sol.** (b)

4. (1 Mark) Which of the following is a good strategy for reducing the variance in a decision tree?

- (a) If improvement of taking any split is very small, don't make a split. (Early Stopping)
- (b) Stop splitting a leaf when the number of points is less than a set threshold K.
- (c) Stop splitting all leaves in the decision tree when any one leaf has less than a set threshold K points.
- (d) None of the Above.

**Sol.** (b)

5. (1 Mark) Which of the following statements about multiway splits in decision trees with categorical features is correct?
- (a) They always result in deeper trees compared to binary splits
  - (b) They always provide better interpretability than binary splits
  - (c) They can lead to overfitting when dealing with high-cardinality categorical features
  - (d) They are computationally less expensive than binary splits for all categorical features

**Sol.** (c)

6. (1 Mark) Which of the following statements about imputation in data preprocessing is most accurate?
- (a) Mean imputation is always the best method for handling missing numerical data
  - (b) Imputation should always be performed after splitting the data into training and test sets
  - (c) Missing data is best handled by simply removing all rows with any missing values
  - (d) Multiple imputation typically produces less biased estimates than single imputation methods

**Sol.** (d)

7. (2 Marks) Consider the following dataset:

| <i>feature1</i> | <i>feature2</i> | <i>output</i> |
|-----------------|-----------------|---------------|
| 18.3            | 187.6           | a             |
| 14.7            | 184.9           | a             |
| 19.4            | 193.3           | a             |
| 17.9            | 180.5           | a             |
| 19.1            | 189.1           | a             |
| 17.6            | 191.9           | b             |
| 19.9            | 190.2           | b             |
| 17.3            | 198.6           | b             |
| 18.7            | 182.6           | b             |
| 15.2            | 187.3           | b             |

Which among the following split-points for *feature2* would give the best split according to the misclassification error?

(a) 186.5

(b) 188.6

(c) 189.2

(d) 198.1

**Sol.** (c)